ThreadMask: thin-filament refinement

Research release report | 16 October 2026

Finding

A full-resolution ridge regressor makes a small improvement over the frozen pixel/component baseline on reused development data. The combined evaluation gains one annotation-level thin match and removes three false positives. This does not establish an independent performance gain or a new competition score.

Evaluation	PQ: baseline / ridge	FP: baseline / ridge	Thin matches: baseline / ridge
106 images / 95 dates	0.271485 / 0.272835	550 / 548	82/314 / 83/314
40 images / 40 dates	0.285986 / 0.287015	260 / 259	44/128 / 44/128
Combined: 146 / 135	0.276155 / 0.277402	810 / 807	126/442 / 127/442

Method

Ten image features include intensity, local dark contrast and dark-ridge Hessian responses at scales 1, 2 and 4 on original 2048 x 2048 images. An ExtraTrees regressor (48 trees, depth 18, minimum leaf 8) learns mean foreground votes from 335,163 sampled pixels across 60 training images. Training oversamples thin foreground; inference uses image pixels only.

The refiner modifies existing instance masks without adding instance IDs. Width 16, probability threshold 0.2 and radius 0 were selected on 19 tuning images. Area/support guards preserve instances with weak support. The same frozen model and settings were then applied to the additional 40 dates.

Interpretation and limitations

The metric is evaluated separately for each annotation set. Thin matches are annotation-level detections, not unique physical filaments. The evaluation-only thin proxy uses 2 x area / inner-boundary pixel count at most 8, with longest bounding-box span at least 40 pixels.

The additional 40 dates were excluded from the current pipeline's training, tuning and first assessment splits. Older baseline experiments had used the available dataset. These are development results, not a blind holdout. The additional set adds no thin matches. Descriptive date-bootstrap intervals include zero: first-assessment PQ gain [-0.001377, 0.004161], additional-set gain [-0.000574, 0.003717].

Reproduction and checks

README.md and REPRODUCE.md describe pinned dependencies, local prediction with separately supplied weights, and preparing a new training workspace. predict.py checks model hashes, accepts original-resolution grayscale input and refuses to overwrite output. A reference-image run is compared against the frozen research label-array digest. The root test suite checks ridge behavior and release integrity. Full training was not repeated during release preparation.

Evidence and distribution

Frozen protocols, tuning outcomes, failed alternatives and aggregate results are retained under outputs/solar-research-v5, solar-thin-v6, solar-thin-v7, solar-ridge-v8 and solar-ridge-validation-v9. Root RELEASE_MANIFEST.json identifies this source-only release; VALIDATION.json records preparation checks. Earlier manifests describe historical full runs.

Original code: Apache-2.0. Competition images, annotations, trained weights, cached predictions and submission CSVs are excluded. Obtain inputs through their authorized source and observe their terms. No result is uploaded and no existing submission is changed. The observed local improvement does not substantiate a prize claim.